

Now $\cos \theta = 1 - 2(\sin \theta/2)^2$, so setting $x = \sin(\theta/2)$ we see that the above integral becomes

$$\frac{\mu^{1/2} e^{2s}}{\pi} \int_{-\frac{\sqrt{2}}{2}}^{\frac{\sqrt{2}}{2}} e^{-4sx^2} dx.$$

However

$$\begin{aligned} \int_{-\frac{\sqrt{2}}{2}}^{\frac{\sqrt{2}}{2}} e^{-4sx^2} dx &= \int_{-\infty}^{\infty} e^{-4sx^2} dx + O\left(\int_{\frac{\sqrt{2}}{2}}^{\infty} e^{-4sx^2} dx\right) \\ &= \frac{\sqrt{\pi}}{2s^{1/2}} + O(e^{-2s}), \end{aligned}$$

and also

$$\frac{d}{ds} \left(\int_{-\frac{\sqrt{2}}{2}}^{\frac{\sqrt{2}}{2}} e^{-4sx^2} dx \right) = \frac{d}{ds} \left(\frac{\sqrt{\pi}}{2s^{1/2}} \right) + O(e^{-2s}).$$

Gathering all the error terms together, we find

$$p(n) = \frac{d}{dn} \left(\mu^{1/2} \frac{e^{2s}}{\pi} \frac{\sqrt{\pi}}{2s^{1/2}} \right) + O(e^{\frac{K}{2}n^{1/2}}).$$

Since $s = \frac{\pi}{\sqrt{6}}(n - \frac{1}{24})^{1/2}$, $\mu = \frac{\sqrt{6}}{12}(n - \frac{1}{24})^{-1/2}$, and $K = \pi\sqrt{\frac{2}{3}}$, this is

$$p(n) = \frac{1}{2\pi\sqrt{2}} \frac{d}{dn} \left(\frac{e^{K(n-\frac{1}{24})^{1/2}}}{(n - \frac{1}{24})^{1/2}} \right) + O(e^{\frac{K}{2}n^{1/2}}),$$

and the theorem is proved.

5 Problems

1. Let η be an indefinitely differentiable function supported in a finite interval, so that $\eta(x) = 1$ for x near 0. Then, if $m > -1$ and $N > 0$,

$$\int_0^\infty e^{-sx} x^m \eta(x) dx = s^{-m-1} \Gamma(m+1) + O(s^{-N})$$

for $\operatorname{Re}(s) \geq 0$, $|s| \rightarrow \infty$.

(a) Consider first the case $-1 < m \leq 0$. It suffices to see that

$$\int_0^\infty e^{-sx} x^m (1 - \eta(x)) dx = O(s^{-N}),$$

and this can be done by repeated integration by parts since $e^{-sx} = (-1)^N s^{-N} \left(\frac{d}{dx}\right)^N (e^{-sx})$.

(b) To extend this to all m , find an integer k so that $k - 1 < m \leq k$, write

$$\int \left[\left(\frac{d}{dx} \right)^k (x^m) \right] e^{-sx} \eta(x) dx = c_{k,m} s^{-m+k-1} + O(s^{-N}),$$

and integrate by parts k times.

2. The following is a more precise version of Stirling's formula. There are real constant $a_1 = 1/12$, $a_2, \dots, a_n, \dots$, so that for every $N > 0$

$$\Gamma(s) = e^{s \log s} e^{-s} \frac{\sqrt{2\pi}}{s^{1/2}} \left(1 + \sum_{j=1}^N a_j s^{-j} + O(s^{-N}) \right) \quad \text{when } s \in S_\delta.$$

This can be proved by using the results of Problem 1 in place of Proposition 1.3.

3. The Bessel functions and Airy function have the following power series expansions:

$$J_\nu(x) = \left(\frac{x}{2} \right)^\nu \sum_{m=0}^{\infty} \frac{(-1)^m \left(\frac{x^2}{4} \right)^m}{m! \Gamma(\nu + m + 1)},$$

$$\text{Ai}(-x) = \frac{1}{\pi} \sum_{n=0}^{\infty} \frac{x^n}{n!} \sin(2\pi(n+1)/3) 3^{n/3-2/3} \Gamma(n/3 + 1/3).$$

(a) From this, verify that when $x > 0$,

$$\text{Ai}(-x) = \frac{x^{1/2}}{3} \left(J_{1/3} \left(\frac{2}{3} x^{3/2} \right) + J_{-1/3} \left(\frac{2}{3} x^{3/2} \right) \right).$$

(b) The function $\text{Ai}(x)$ extends to an entire function of order $3/2$.

[Hint: For (b), use (a), or alternatively, apply Problem 4 in Chapter 5 to the power series for Ai . Compare also with Problem 1, Chapter 4.]

4. Suppose $F(z) = \sum_{n=0}^{\infty} F_n w^n$ is meromorphic in a region containing the closed unit disc, and the only poles of F are on the unit circle at the points $\alpha_1, \dots, \alpha_k$, and their orders are $r_1, \dots, r_k$ respectively. Then for some $\epsilon > 0$

$$F_n = \sum_{j=1}^k P_j(n) + O(e^{-\epsilon n}) \quad \text{as } n \rightarrow \infty.$$

Here

$$P_j(n) = \frac{1}{(r_j - 1)!} \left(\frac{d}{dw} \right)^{r_j - 1} [(w - \alpha_j)^{r_j} w^{-n-1} F(w)]_{w=\alpha_j}.$$

Note that each P_j is of the form $P_j(n) = A_j(\alpha_j^{-1}n)^{r_j-1} + O(n^{r_j-2})$.

To prove this, use the residue formula (Theorem 1.4, Chapter 3).

5.* The one shortcoming in our derivation of the asymptotic formula for $p(n)$ arose from the fact that while $f_1(z) = \sqrt{z/i} e^{\frac{i\pi}{12z}} e^{\frac{i\pi z}{12}}$ is a good approximation to the generating function $f(z)$ near $z = 0$, this fails near other points on the real axis, since f_1 is regular there, but f is not.

However, using the transformation law (24) and the identity $f(z+1) = f(z)$, one can derive the following generalization of (24): whenever p/q is a rational number in lowest form (so p and q are relatively prime) then

$$f\left(z - \frac{p}{q}\right) = \omega_{p/q} \sqrt{\frac{zq}{i}} e^{\frac{i\pi}{12zq^2}} e^{-\frac{i\pi z}{12}} f\left(-\frac{1}{zq^2} - \frac{p'}{q}\right),$$

where $pp' = 1 \pmod{q}$. Here $\omega_{p/q}$ is an appropriate 24^{th} root of unity. This formula leads to an analogous $f_{p/q}$, approximating f at $z = p/q$.

From this one can obtain for each p/q a contribution of the form

$$c_{p/q} \frac{1}{2\pi\sqrt{2}} \frac{d}{dn} \left(\frac{e^{\frac{K}{q}(n-\frac{1}{24})^{1/2}}}{(n-\frac{1}{24})^{1/2}} \right)$$

to the asymptotic formula for $p(n)$. When suitably modified, the resulting series, summed over all proper fractions p/q in $[0, 1)$, actually converges and gives an exact formula for $p(n)$.

Appendix B: Simple Connectivity and Jordan Curve Theorem

Jordan was one of the precursors of the theory of functions of a real variable. He introduced in this part of analysis the capital notion of functions of bounded variation. Not less celebrated is his study of curves, universally called Jordan curves, which curves separate the plane in two distinct regions. We also owe him important propositions regarding the measure of sets that have led the way to numerous modern researches.

E. Picard, 1922

The notion of simple connectivity is at the source of many basic and fundamental results in complex analysis. To clarify the meaning of this important concept, we have gathered in this appendix some further insights into the properties of simply connected sets. Closely tied to the idea of simple connectivity is the notion of the “interior” of a simple closed curve. The theorem of Jordan states that this interior is well-defined and is simply connected. We prove here the special case of this theorem for curves which are piecewise-smooth.

Recall the definition in Chapter 3, according to which a region Ω is simply connected if any two curves in Ω with the same end-points are homotopic. From this definition we deduced an important version of Cauchy’s theorem which states that if Ω is simply connected and $\gamma \subset \Omega$ is any closed curve, then

$$(1) \quad \int_{\gamma} f(\zeta) d\zeta = 0$$

whenever f is holomorphic in Ω . Here, we shall prove that a converse also holds, therefore:

- (I) A region Ω is simply connected if and only if it is holomorphically simply connected; that is, whenever $\gamma \subset \Omega$ is closed and f holomorphic in Ω then (1) holds.

Besides this fundamental equivalence, which is analytic in nature, there